Computer Vision

Multi-Label Image Classification
Query2Label & MLSPL

Group 25
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Background

Loss functions

- Measures distance between prediction (model output) and ground truth.
- Guides parameter optimization.

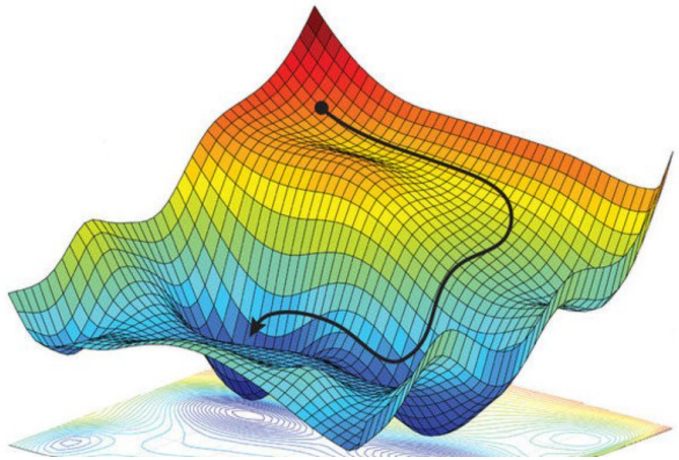


Illustration of a loss landscape.

Background

Loss functions covered:

- Binary Cross-Entropy (BCE)
- Expected Positive Regularisation (EPR)
- Regularised Online Label Estimation (ROLE)

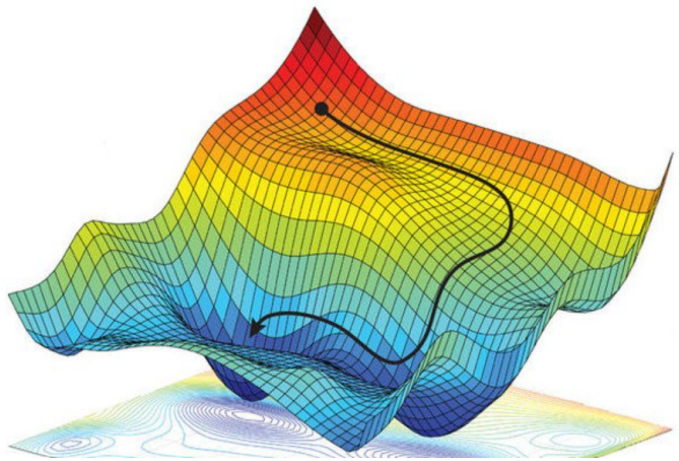
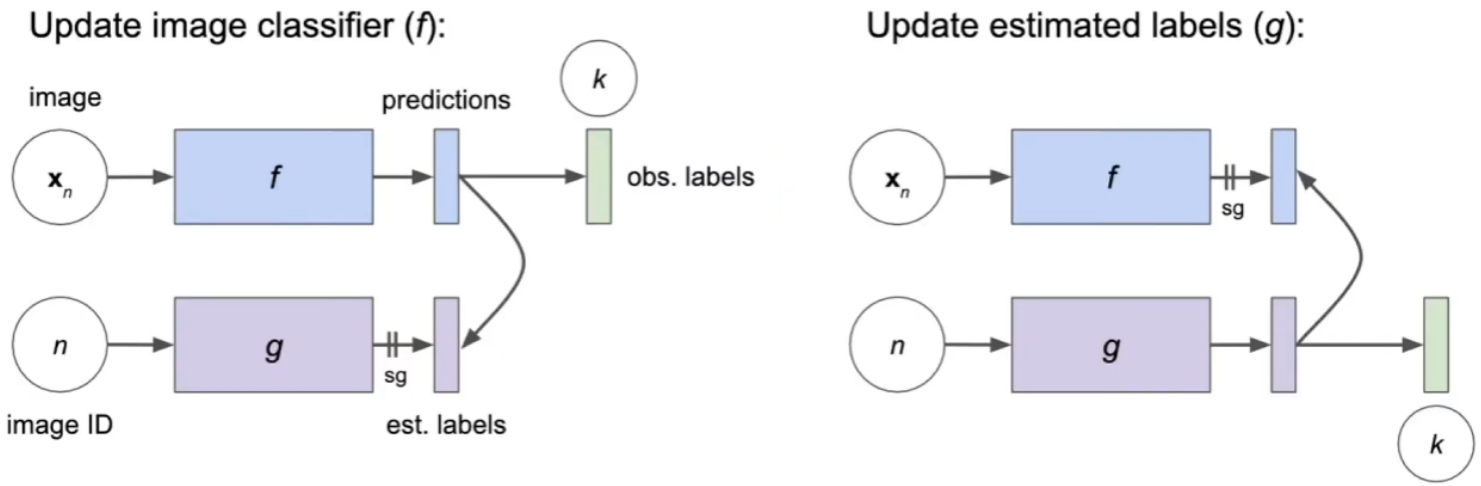


Illustration of a loss landscape.

Regularised Online Label Estimation (ROLE)



Alternating updates of image classifier (f) and label estimator (g).

Background

Binary Cross-Entropy (BCE):

- Independent positive/negative prediction per class: $y_i \in \{0, 1\}$
- Model outputs probability $p_i \in [0, 1]$ for each class $i \in \{1, \ldots, K\}$.

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{K} \sum_{i=1}^{K} [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

Background

BCE limitations:

- **Class imbalance:** gradients dominated by frequent classes $\rightarrow$ rare classes under-represented.
- **Missing labels:** datasets rarely fully annotated. BCE treats unobserved labels as negatives, introducing false-negative bias.

Positive-only BCE used when only the observed positive label $z_{ni} = 1$ is known:

$$\mathcal{L}^{+}_{\text{BCE}}(\mathbf{f}_n, \mathbf{z}_n) = -\sum_{i=1}^{K} \mathbf{1}[z_{ni} = 1] \log f_{ni}$$

Background

Expected Positive Regularisation (EPR)

$$\mathcal{L}_{EPR}(\mathbf{F}_B, \mathbf{Z}_B) = \frac{1}{|B|} \sum_{n \in B} \mathcal{L}^{+}_{BCE}(\mathbf{f}_n, \mathbf{z}_n) + \lambda R_\kappa(\mathbf{F}_B),$$

- κ: expected positives per image (domain prior).
- $\hat{\kappa}$: average predicted positives in the batch.
- Batch penalty: $R_\kappa(\mathbf{F}_B) = \left(\frac{\hat{\kappa}(\mathbf{F}_B) - \kappa}{K}\right)^2$.

Regularised Online Label Estimation (ROLE)

- Jointly trains classifier $f(\cdot;\theta)$ and label-estimator $g(\cdot;\phi)$.
- Alternating updates:

$$\mathcal{L}'(\mathbf{F}_B, \tilde{\mathbf{Y}}_B) = \frac{1}{|B|} \sum_{n \in B} \mathcal{L}_{BCE}(\mathbf{f}_n, \text{sg}(\tilde{\mathbf{y}}_n)) + \mathcal{L}_{EPR}(\mathbf{F}_B, \mathbf{Z}_B)$$

$\mathbf{F} \in [0,1]^{N \times K}$: matrix of classifier predictions.
$\tilde{\mathbf{Y}} \in [0,1]^{N \times K}$: matrix of estimated labels.
$\mathbf{Z}_B$: fixed partial labels.

Final equation:

$$\mathcal{L}_{ROLE}(\mathbf{F}_B, \tilde{\mathbf{Y}}_B) = \frac{\mathcal{L}'(\mathbf{F}_B|\tilde{\mathbf{Y}}) + \mathcal{L}'(\tilde{\mathbf{Y}}|\mathbf{F}_B)}{2}$$

Experiments

- **Hardware:** 2 × NVIDIA RTX 3060 GPUs (12 GB VRAM) on Ubuntu 22.04.5 LTS.
- **Q2L Environment:**
  - Python 3.7.3, PyTorch 1.9.0, Torchvision 0.10.0, CUDA 11.1.
  - `inplace_abn` rebuilt from earlier release for compatibility.
  - **RandAugment dependency missing:**
    - Added local `augmentations.py` from *pytorch-randaugment*.
    - Changed import in `get_dataset.py`: `from .augmentations import RandAugment`.
    - Replaced default call with `RandAugment(n=2, m=9)` to set parameters explicitly.
- **MLSPL Environment:**
  - Python 3.11.8, PyTorch 2.2.1, Torchvision 0.17.1, CUDA 12.4.
  - Single-GPU training to ensure reproducibility across runs.

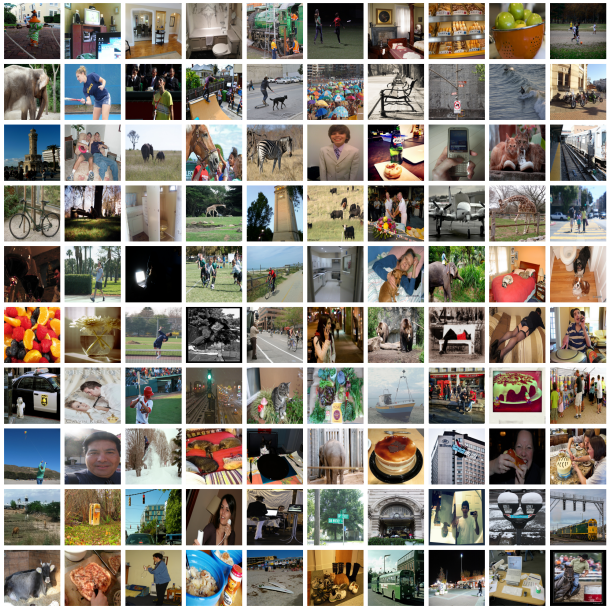
Experiments

Dataset: MS-COCO 2014

- 82,081 training images
- 40,137 validation images.
- 80 categories - multiple objects per image.

Metric: mean Average Precision (mAP)

- Widely reported metric in MLC.
- Used in both Q2L and MLSPL.



Example COCO images (448×448)

Experiments

Q2L Backbones evaluated

- ResNet-101: 448 and 576
- TResNetL 448
- TresNetL (22k) 448
- Swin-L (22k) 384
- CvT-w24 (22k) 384

Training protocol

- Pre-trained weights from authors - batch size 16.
- Scratch training attempted with all backbones.
- Memory overflow with SwinL and CvT-w24 during training.

Experiments

MLSPL Experiments

- Convert each image to exactly one positive label.
- *Linear* classifier on fixed features (25 epochs).
- End-to-end *fine-tuning* (10 epochs).

Hyperparameter search

- Learning rate $\in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$.
- Batch size $\in \{8, 16\}$.

Selected configuration

- **Linear**: lr $= 10^{-3}$, batchsize 16.
- **Fine-tuned**: lr $= 10^{-5}$, batchsize 16.

Q2L

Table 1: Mean Average Precision (mAP) comparison between our reproduced results and the Q2L paper on MS-COCO 2014. All values in percentage.

Method	Backbone	Resolution	mAP (Ours)	mAP (Paper)
Q2L-R101	ResNet-101	448×448	84.9	84.9
Q2L-R101	ResNet-101	576×576	86.5	86.5
Q2L-TresL	TResNetL	448×448	87.3	87.3
Q2L-Tres	TResNetL(22k)	448×448	89.2	89.2
Q2L-SwinL	Swin-L(22k)	384×384	90.5	90.5
Q2L-CvT	CvT-w24(22k)	384×384	91.3	91.3

Result

MLSPL

Table 2: Comparison of mean Average Precision (mAP) results between our implementation and the originally reported values for the MLSPL $\mathcal{L}_{ROLE}$ method on the MS-COCO 2014 dataset. Results are shown for both linear and fine-tuned variants. All mAP values are reported in percentage.

Loss	Method	mAP(Ours)	mAP (Paper)
$\mathcal{L}_{ROLE}$	Linear	66.3	66.3
$\mathcal{L}_{ROLE}$	Fine-Tuned	66.9	66.3